

WASP-52b

R-0535

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_240901 · created 2026-07-21T20:38:02+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460554.8317 +/- 0.0081 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.214 +/- 0.062
Transit depth (Rp/Rs)^2	4.58 +/- 2.65 [%]
Orbital Inclination (inc)	89.5 +/- 2.78
Airmass coefficient 1 (a1)	0.031 +/- 0.019
Airmass coefficient 2 (a2)	2.93 +/- 1.44
Scatter in the residuals of the lightcurve fit is	61.26 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	3.0
Transit Duration (day)	0.0832 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.214 ± 0.062 vs 0.1646 (deviation 0.8σ)
Duration fit vs published	0.0832 d vs 0.0758 d (deviation 0.74σ)
Mid-time O–C	-6.2 min
Depth SNR	3.5
Residual scatter	61.26 %

Light curve

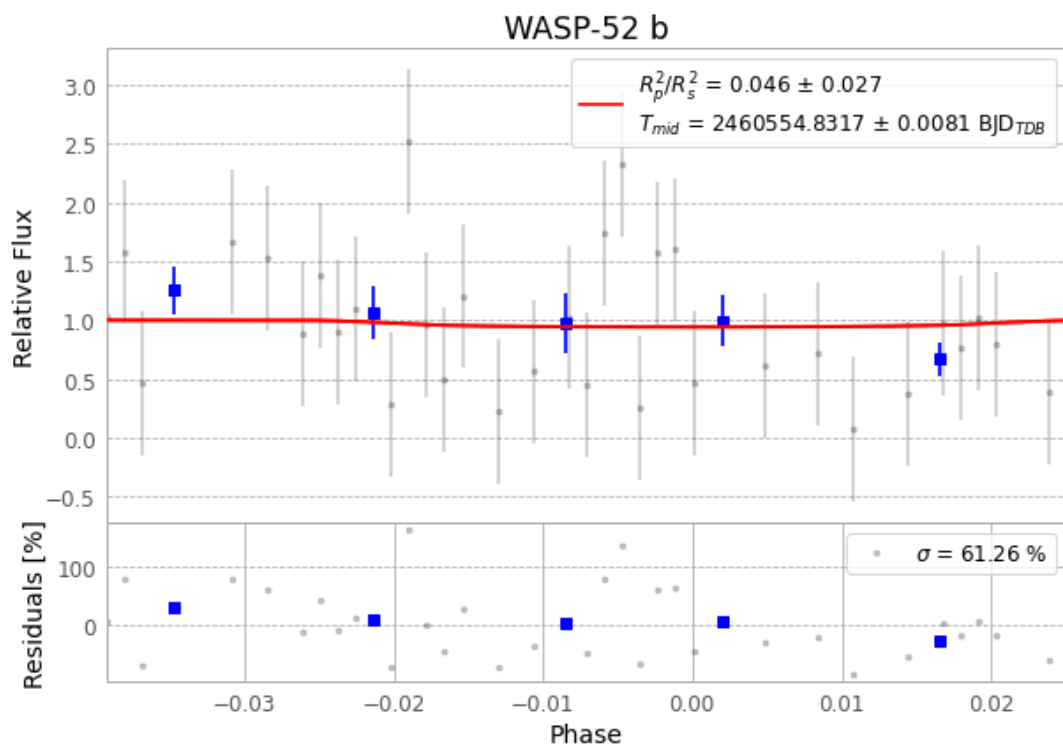


Figure 1 — FinalLightCurve_WASP-52 b_2024-08-31.png (SHA-256 fac4201113ea98fc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [568, 52], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3a72b6592f83f5f9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

55 FITS frames (36 MB), WASP-52240901060915.FITS → WASP-52240901085116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-240901.log (9fcbb0ec2442...), FinalLightCurve_WASP-52 b_2024-08-31.png (fac4201113ea...), FinalLightCurve_WASP-52 b_2024-08-31.csv (b8c464941f1d...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 20:38Z.